Seeing chiral superconductivity in QPI: results

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I. SETUP

Nearest-neighbour triangular lattice, single impurity, broadening $\Gamma = 10^{-3}$, $V_0 = 0.1$, $|\Delta| = 0.1$ (units of the hopping), 240×240 $\boldsymbol{k}$-mesh, 120×120 $\boldsymbol{q}$-mesh. Spinful Nambu basis $(c_{\boldsymbol{k}\uparrow}, c_{\boldsymbol{k}\downarrow}, c^\dagger_{-\boldsymbol{k}\uparrow}, c^\dagger_{-\boldsymbol{k}\downarrow})$; the LDOS is resolved by spin, $\delta\rho_\mu$ with $\mu \in \{0, x, y, z\}$. Born approximation unless stated, with a Gaussian momentum cutoff e^{-q^2/q_0^2} (q_0 in units of $|\boldsymbol{b}_1|$). Every $\delta\rho(\boldsymbol{q})$ is projected onto the irreps of C_{6v}, with weight $\sum |f_\nu|^2 / \sum |f|^2$. Pairings: s, extended s, $d_{x^2-y^2}$ (nematic) and $d+id$ (chiral), singlet, at $\omega = 0$ and ω_2 just below the gap edge ($\omega_2 = 0.15$ for the nodal $d_{x^2-y^2}$ state). Each panel is normalised to its own maximum, given below it. Table I summarises all cases.

II. NO SPIN-ORBIT COUPLING

a. Scalar (A_1) impurity. All spin channels vanish. $\delta\rho$ is purely A_1 for s, extended s and $d+id$; only the nematic $d_{x^2-y^2}$ state has an E_2 part (Fig. 1). A scalar impurity sees nematicity but not chirality.

b. A_2 impurity. $V(\boldsymbol{k}, \boldsymbol{k}') = iV_0(\boldsymbol{k}\times\boldsymbol{k}')_z\, e^{-|\boldsymbol{k}-\boldsymbol{k}'|^2/q_0^2}$, $q_0 = 20|\boldsymbol{b}_1|$. $\delta\rho$ vanishes for every case except $d+id$ at ω_2, where it is purely A_1 (Fig. 2).

III. RASHBA SOC, ANSATZ PAIRINGS

$\hat{h}(\boldsymbol{k}) = \varepsilon(\boldsymbol{k}) + \alpha\,(\partial_{k_y}\varepsilon\,\sigma_x - \partial_{k_x}\varepsilon\,\sigma_y)$, $\alpha = 0.1$, $\mu = -0.1$, $\hat{\Delta} = \Psi(\boldsymbol{k})\, i\sigma_y$; scalar impurity. Bands in Fig. 3. For s, extended s and $d_{x^2-y^2}$ only $\delta\rho_0$ is finite (Fig. 5). For $d+id$ at ω_2, all four channels are finite: $\delta\rho_0, \delta\rho_z \in A_1$ and $\mathrm{Im}\,\delta\rho_{x,y} \in E_1$ (Fig. 4). An A_1 pattern in $\delta\rho_z$ requires broken time reversal and broken mirrors together, i.e. the chiral state.

IV. RASHBA-HUBBARD MODEL, FRG PAIRING

Leading $\boldsymbol{q} = 0$ pairing instability of a divERGe FRG flow at half filling, $\Delta_{\mathrm{rms}} = 0.1$; scalar impurity. $U = 0$, $\alpha = 0.1$ (non-degenerate): only $\delta\rho_0 \in A_1$ (Fig. 7). $U = 8$, $\alpha = 0.4$ (two-fold degenerate; chiral combination $(v_0 + iv_1)/\sqrt{2}$): the ansatz $d+id$ pattern, already finite at $\omega = 0$ (Fig. 6).

V. FULL T-MATRIX

Point scalar impurity, no momentum cutoff, $\hat{T}(\omega) = [1 - \hat{V}\hat{G}_{\mathrm{loc}}(\omega)]^{-1}\hat{V}$ with $\hat{V} = V_0\tau_z$, at $V_0 = 0.1$ and at the resonant strengths $V_0 = 1/\lambda$ (λ a real eigenvalue of $\hat{G}_{\mathrm{loc}}(\omega, \Gamma = 0)\,\tau_z$; Table II). The full T-matrix never adds an irrep to a channel; it only changes relative amplitudes. The one change to Table I is Rashba $d+id$ at $\omega = 0$, where the spin channels become finite.

TABLE I. Irreps of the spin-resolved QPI $\delta\rho_\mu(\boldsymbol{q})$ for each system, impurity and pairing. 0 means the channel vanishes. Each pairing is labelled by its own C_{6v} irrep, which for FRG includes spin-symmetric pairings allowed under Rashba SOC. Each entry combines $\omega = 0$ and ω_2. All A_1-irrep impurities hold both in the Born approximation and with the full T-matrix. The † notes where the QPI is instead zero in the Born approximation at $\omega = 0$. The A_2 impurity case was only calculated in the Born approximation. "Im" marks a channel that is purely imaginary in $\boldsymbol{q}$-space; all others are real. Irreps with weight below 10^{-3} are omitted. The last column gives the figures in which that case's QPI is plotted.

System	Impurity	Pairing	ω_2	$\delta\rho_0$	$\delta\rho_x$	$\delta\rho_y$	$\delta\rho_z$	Fig.
no SOC	A_1	A_1: s	0.0980	A_1	0	0	0	1(a)
		A_1: ext. s	0.0098	A_1	0	0	0	1(b)
		E_2: $d + id$	0.2142	A_1	0	0	0	1(c)
		E_2: $d_{x^2-y^2}$	0.1500	A_1 + E_2	0	0	0	1(d)
	A_2	A_1: s	0.0980	0	0	0	0	–
		A_1: ext. s	0.0098	0	0	0	0	–
		E_2: $d + id$	0.2142	$A_1{}^\dagger$	0	0	0	2
		E_2: $d_{x^2-y^2}$	0.1500	0	0	0	0	–
Rashba	A_1	A_1: s	0.0980	A_1	0	0	0	5(a)
		A_1: ext. s	0.0187	A_1	0	0	0	5(b)
		E_2: $d + id$	0.2095	A_1	Im: $E_1{}^\dagger$	Im: $E_1{}^\dagger$	$A_1{}^\dagger$	4
		E_2: $d_{x^2-y^2}$	0.1500	A_1 + E_2	0	0	0	5(c)
Rashba–Hubbard	A_1	A_1: U=0, V=0.9	0.0761	A_1	0	0	0	7
(FRG)		E_2: U=8, V=0, χ=+1	0.0846	A_1	Im: E_1	Im: E_1	A_1	6

TABLE II. Full T-matrix at the resonant impurity strengths $V_0 = 1/\lambda$, scalar impurity. $\max|\hat{T}|$, at $\Gamma = 10^{-3}$, measures how sharp the resonance is. Ratios are of the $\boldsymbol{q}$-space maxima, including $\boldsymbol{q} = 0$; 0 means numerically zero. In every row $\delta\rho_0, \delta\rho_z \in A_1$ and $\mathrm{Im}\,\delta\rho_{x,y} \in E_1$, each with weight 1, where finite. Only states with SOC were searched. s pairing and the $U = 0$ FRG state have no resonances; the nodal $d_{x^2-y^2}$ state has no gap, so resonances are not considered.

State	ω	V_0	$\max\|\hat{T}\|$	$\delta\rho_z/\delta\rho_0$	$\delta\rho_{x,y}/\delta\rho_0$
$d + id$ (Rashba)	0.0000	6.28	783	3.7×10^{-3}	6.7×10^{-3}
	0.2095	−2.28	36.8	1.0×10^{-4}	2.4×10^{-4}
	0.2095	1.33	40	1.3×10^{-2}	3.8×10^{-2}
ext. s (Rashba)	0.0187	−2.96	5.76	0	0
	0.0187	1.60	9.29	0	0
FRG $U = 8$, $\chi = +1$	0.0000	5.38	262	4.0×10^{-2}	5.6×10^{-2}
	0.0000	6.56	262	4.1×10^{-2}	5.2×10^{-2}
	0.0846	−1.96	9.54	5.3×10^{-2}	5.4×10^{-2}
	0.0846	−1.72	8.86	3.7×10^{-2}	1.5×10^{-2}
	0.0846	1.12	10.9	3.7×10^{-2}	6.2×10^{-2}
	0.0846	1.22	11.9	7.0×10^{-2}	6.9×10^{-2}

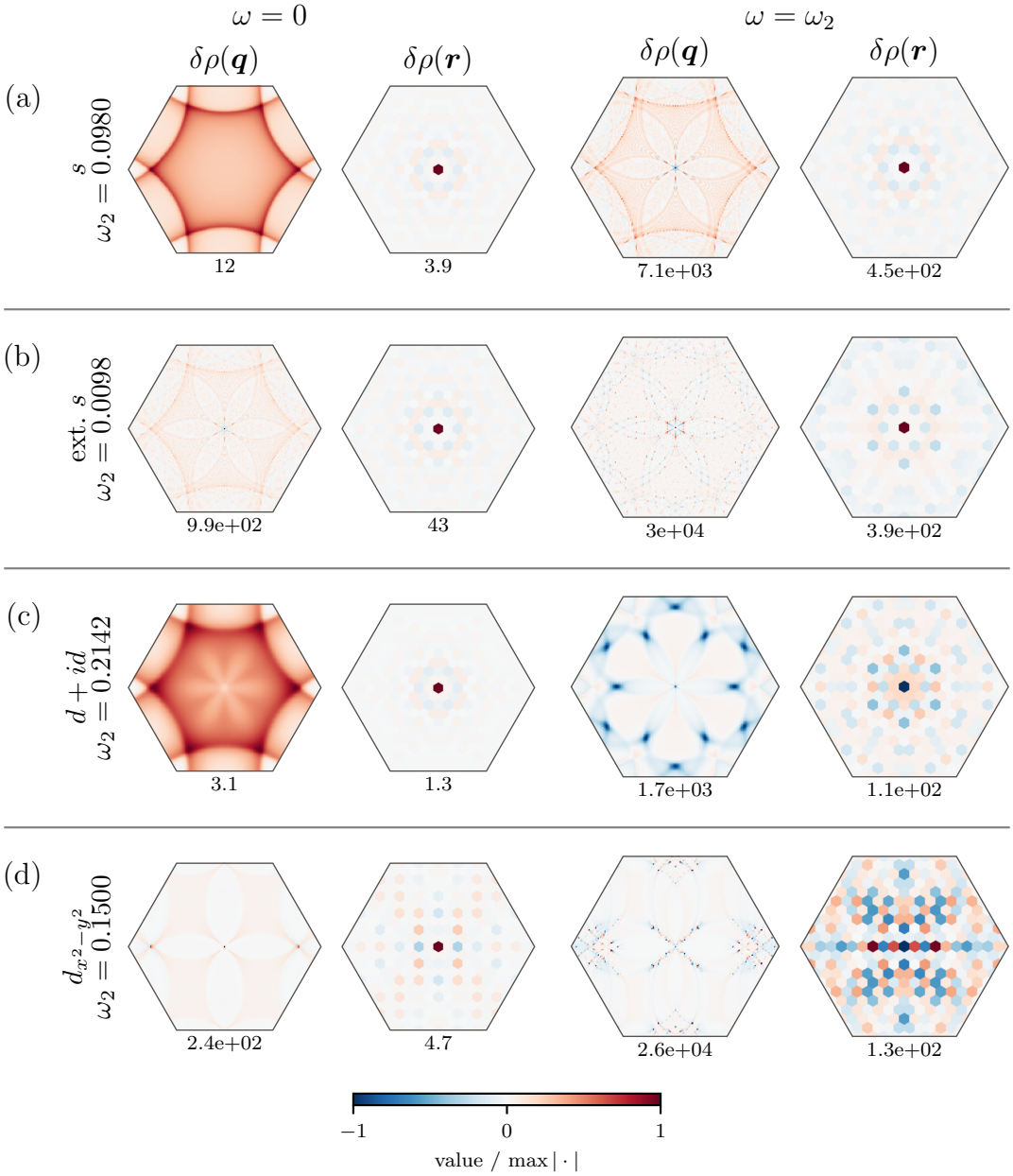


FIG. 1. No SOC, scalar (A_1) impurity, $q_0 = |\boldsymbol{b}_1|$. Rows (a)–(d): s, extended s, $d + id$ and $d_{x^2-y^2}$ pairing. Columns: $\mathrm{Re}\,\delta\rho(\boldsymbol{q})$ and $\delta\rho(\boldsymbol{r})$ at $\omega = 0$, then the same at ω_2, which is given on each row. $\mathrm{Im}\,\delta\rho(\boldsymbol{q})$ and all spin channels vanish. Only the nematic $d_{x^2-y^2}$ state has a component beyond A_1, from the E_2 irrep, which breaks C_6 down to C_2.

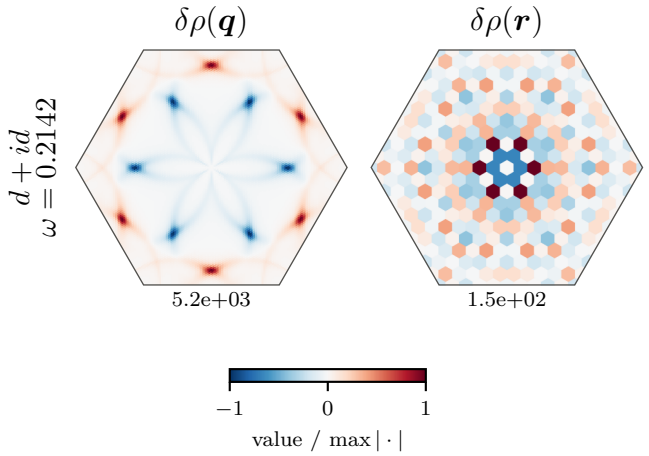


FIG. 2. No SOC, A_2 impurity, $q_0 = 20|\boldsymbol{b}_1|$: the only non-zero case, $d + id$ at $\omega_2 = 0.2142$, with the panel conventions of Fig. 1. $\delta\rho$ vanishes identically for s, extended s and $d_{x^2-y^2}$ pairing at both frequencies, and for $d + id$ at $\omega = 0$. The response is purely A_1.

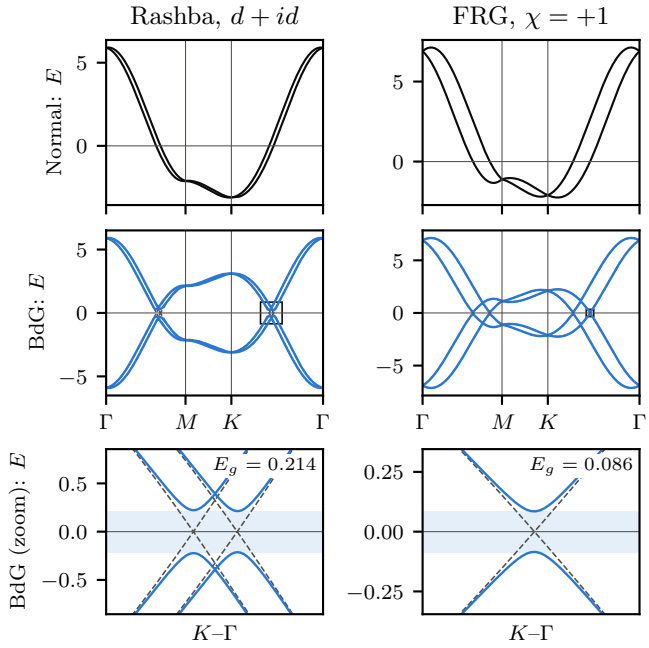


FIG. 3. Bands on Γ–M–K–Γ for the two chiral states. Left: Rashba model ($\alpha = 0.1$, $\mu = -0.1$) with $d+id$ singlet pairing, $|\Delta| = 0.1$. Right: FRG gap of the Rashba-Hubbard model ($U = 8$, $\alpha = 0.4$, half filling), $\chi = +1$, $\Delta_{\mathrm{rms}} = 0.1$. Rows: normal state; BdG bands (solid) with $\pm\varepsilon(\boldsymbol{k})$ (dashed); BdG bands zoomed on the boxed region, the Fermi crossing with the smallest gap on the path. The shading marks $|E| < E_g$, with E_g the gap over the whole Brillouin zone.

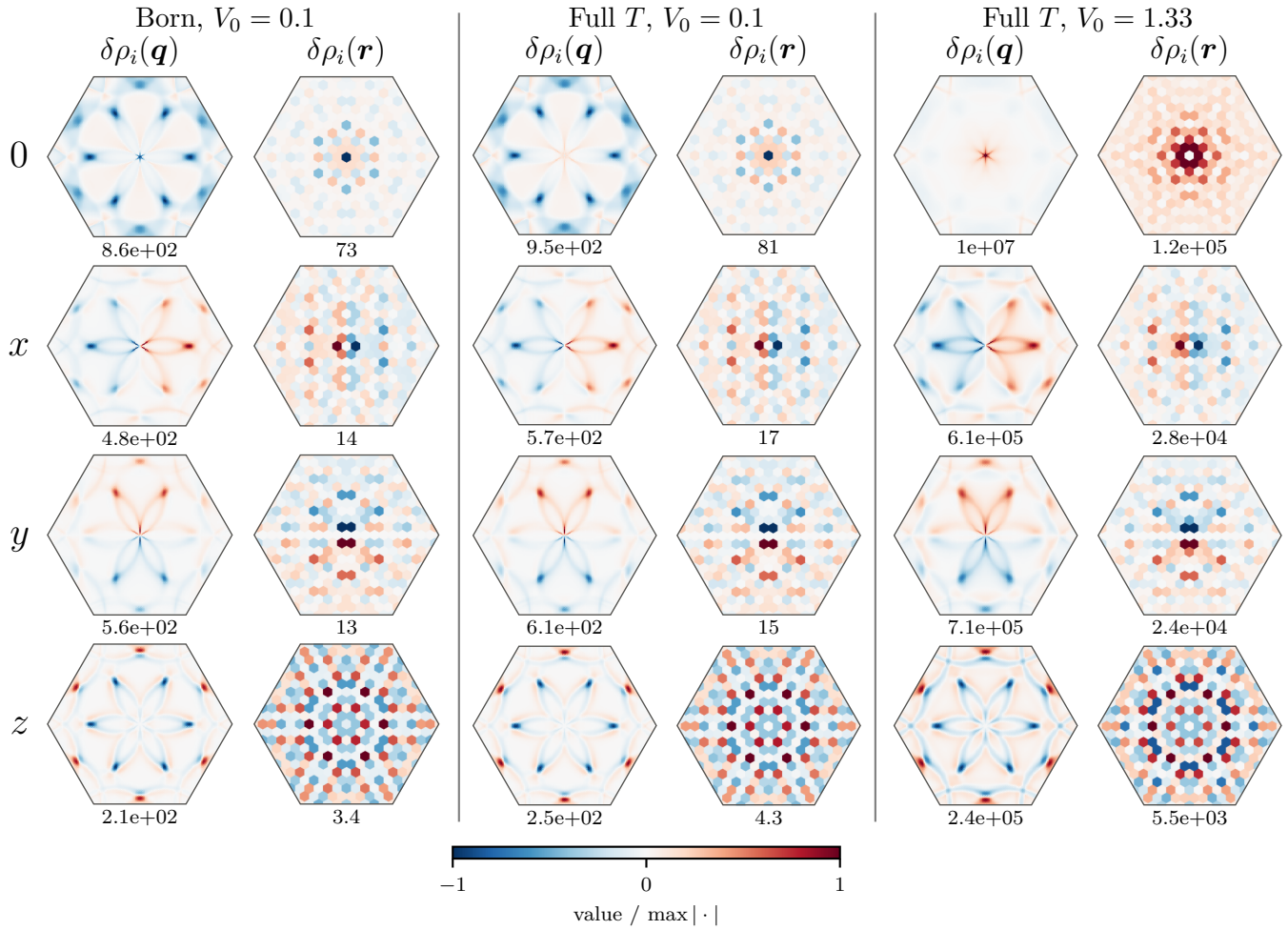


FIG. 4. Spin-resolved QPI for the Rashba model with $d + id$ pairing and a scalar impurity, at $\omega_2 = 0.2095$. Left: Born approximation, $q_0 = |\boldsymbol{b}_1|$. Centre: full T-matrix at the same V_0. Right: full T-matrix at the resonance $V_0 = 1.33$ ($\max|\hat{T}| = 40$). Here $\delta\rho_0(\boldsymbol{q})$ is dominated by a broad peak at $\boldsymbol{q} = 0$ from the impurity bound state, which hides the scattering pattern of the other two columns; the spin channels keep that pattern. The full-T panels have no q cutoff. Rows are $i = 0, x, y, z$; the $\boldsymbol{q}$-maps show $\mathrm{Im}\,\delta\rho_i$ for x, y and $\mathrm{Re}\,\delta\rho_i$ for $0, z$, and the other part vanishes. Each panel is normalised to its own maximum, given below it; for the $\boldsymbol{q}$-maps this is the maximum over $\boldsymbol{q} \neq 0$, so the $\boldsymbol{q} = 0$ point saturates. The irreps are given in Table I.

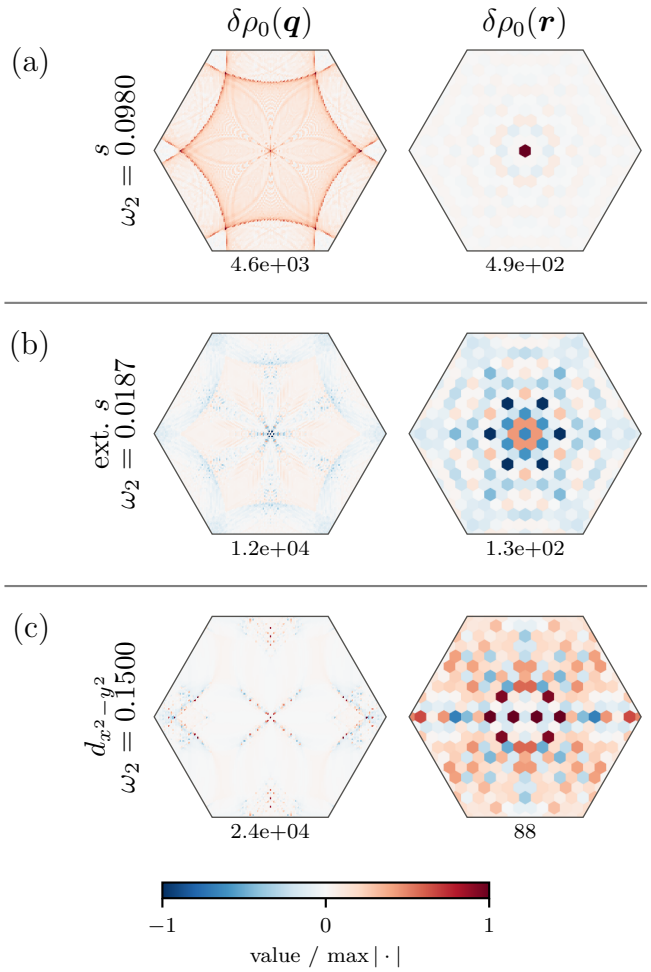


FIG. 5. $\mathrm{Re}\,\delta\rho_0$ for the Rashba model with s, extended s and $d_{x^2-y^2}$ pairing, scalar impurity, Born approximation, $q_0 = |\boldsymbol{b}_1|$, each at its ω_2. All spin channels vanish. Normalisation as in Fig. 4.

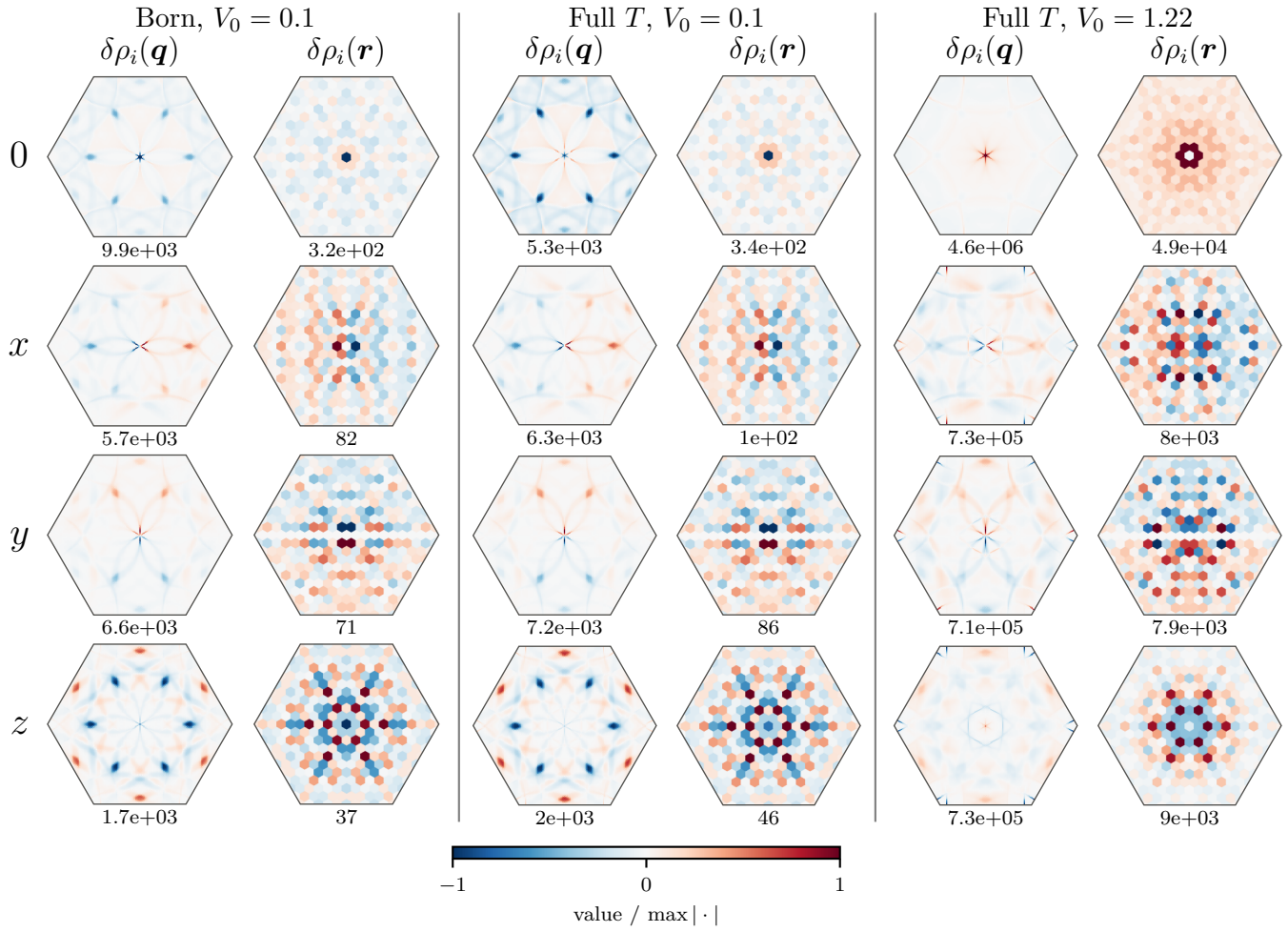


FIG. 6. Spin-resolved QPI for the chiral FRG state of Fig. 3, scalar impurity, $\omega = 0.0846$. Left: Born approximation, $q_0 = |\boldsymbol{b}_1|$. Centre: full T-matrix at the same V_0. Right: full T-matrix at the resonance $V_0 = 1.22$ ($\max|\hat{T}| = 12$), the sharpest of the four at this ω; as in Fig. 4, the bound state dominates $\delta\rho_0$, while the spin channels keep their pattern. The full-T panels have no q cutoff. Panel conventions as in Fig. 4; $\delta\rho_z$ is A_1 in all three.

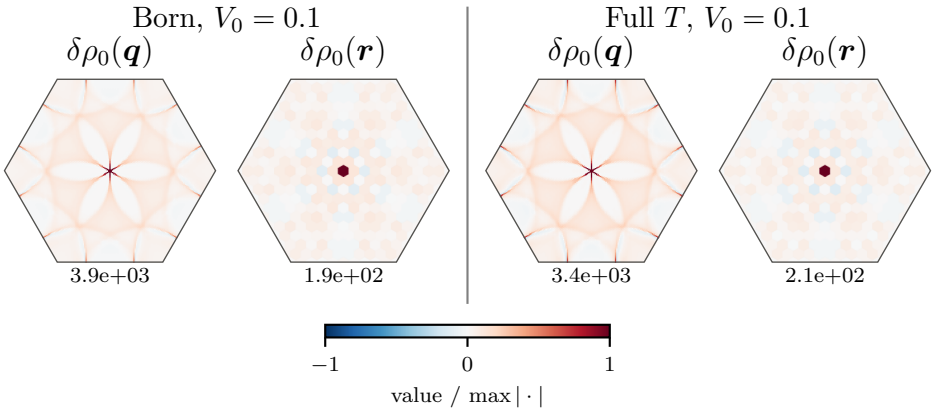


FIG. 7. $\mathrm{Re}\,\delta\rho_0$ for the non-chiral FRG state ($U = 0$, $\alpha = 0.1$), scalar impurity, $\omega = 0.0761$, as in Fig. 6. All spin channels vanish. This state has no impurity bound states, so there is no resonant case.